

Final Business Report

Employee Attrition Analysis

Business Context:

Employee turnover represents a major challenge for organizations because it increases recruitment costs, disrupts team productivity, and results in the loss of valuable organizational knowledge.

This analysis investigates the factors influencing employee attrition using historical HR data. The objective is to identify the characteristics of employees who are most likely to leave the organization and provide data-driven recommendations that improve employee retention.

The dataset indicates that approximately 17.25% of employees left the company, while 82.75% remained.

Executive Findings:

The analysis identified several major drivers of employee attrition.

1.Overtime is the strongest attrition indicator

Employees working overtime consistently exhibit a much higher likelihood of leaving the organization than employees with normal working schedules.

2.Early-career employees represent the highest-risk group

Attrition is concentrated among employees during their early years of employment, particularly:

- Junior employees
- Employees with relatively short company tenure

This suggests that retention efforts should focus on employees before they become fully established within the organization.

3.Lower compensation is associated with higher attrition

Employees who resigned generally earn lower monthly salaries compared with employees who stayed.

Although compensation is not the sole reason for turnover, salary appears to be an important retention factor.

4.Travel requirements increase turnover

Employees who travel frequently for work experience noticeably higher attrition than employees who rarely travel or do not travel.

Frequent travel may contribute to stress, fatigue, and reduced work-life balance.

5.Job satisfaction influences retention

Employees with low job satisfaction leave at considerably higher rates than employees reporting higher satisfaction.

Improving employee experience could therefore reduce future turnover.

6.Marital status shows meaningful differences

Single employees experience the highest attrition among all marital groups, suggesting demographic characteristics may also contribute to resignation patterns.

Business Objectives:

The objectives of this project were to:

- Assess overall employee attrition.
 - Clean and prepare HR data for analysis.
 - Identify the strongest drivers of employee turnover.
 - Generate actionable business recommendations.
 - Prepare the dataset for future machine learning models.
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Dataset Overview:

The HR dataset contains employee demographic information, compensation details, job characteristics, satisfaction metrics, and employment history.

Target Variable

Attrition

- 0 → Employee Stayed
- 1 → Employee Left

Important Variables

• Age	• JobSatisfaction
• MonthlyIncome	• TotalWorkingYears
• OverTime	• YearsAtCompany
• BusinessTravel	• MaritalStatus
• JobLevel	•

Data Preparation:

Several preprocessing steps were performed before the analysis.

Duplicate Removal

Duplicate employee records were identified and removed.

Missing Values:

Missing numerical values were replaced using the median.

Examples include:

- Age
- MonthlyIncome
- TotalWorkingYears
- YearsAtCompany

Missing categorical values were replaced using the mode, including:

- EducationField
 - JobRole
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Data Cleaning:

Categorical values were standardized to eliminate inconsistent spellings and formatting.

Examples include:

- sales → Sales
- Human Resources → Human Resources
- Sales Ex. → Sales Executive

Whitespace and inconsistent capitalization were also corrected.

Removing Irrelevant Columns

Columns containing no analytical value were removed.

Examples include:

- EmployeeCount
 - StandardHours
 - Over18
 - EmployeeNumber
 - RecordID
 - ManagerNotes
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Outlier Treatment:

Extreme observations were removed using the Interquartile Range (IQR) method for:

- Age
- MonthlyIncome
- DistanceFromHome

This reduced the influence of unusually extreme values and improved the reliability of descriptive statistics.

Encoding

Categorical variables were converted into numerical form.

Techniques included:

- Label Encoding
 - One-Hot Encoding
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Feature Scaling

Numerical features were standardized using StandardScaler to prepare the dataset for future machine learning models.

Exploratory Data Analysis

Overall Attrition

Only a minority of employees left the organization.

Approximately 17.25% resigned, while the remaining employees stayed.

1.Overtime

Employees working overtime experience substantially higher attrition than employees who do not work overtime.

This indicates that workload and burnout are likely major contributors to employee turnover.

2.Monthly Income

Employees who resigned generally receive lower monthly salaries.

The salary distribution suggests that lower-paid employees are more likely to leave the organization.

3.Department

Although Research & Development contributes the largest number of resignations, it also employs the largest workforce.

When attrition is evaluated proportionally, Human Resources and Sales experience higher attrition rates than Research & Development.

4.Job Satisfaction

Attrition steadily decreases as job satisfaction increases.

Employees reporting the lowest satisfaction levels are considerably more likely to resign.

5.Age

Younger employees demonstrate the highest turnover.

Employee retention improves noticeably among middle-aged employees.

6.Business Travel

Business travel has a clear relationship with attrition.

Employees who travel frequently are much more likely to leave than employees who rarely travel or never travel.

7.Job Level

Entry-level employees exhibit the highest attrition.

Turnover generally declines as employees progress into higher organizational levels.

8.Training

Employees who received no training during the previous year experienced the highest attrition.

Training appears to support employee engagement and long-term retention.

9.Marital Status

Single employees are considerably more likely to resign than married or divorced employees.

Correlation Analysis

Correlation analysis identified the variables most strongly associated with employee attrition.

The strongest relationships include:

- OverTime
- MonthlyIncome
- JobLevel
- TotalWorkingYears
- YearsAtCompany
- Age

Although individual correlations are moderate, their combined effects reveal meaningful employee risk patterns.

Feature Engineering

Several new variables were created to improve business analysis.

Feature	Purpose
ExperienceLevel	Categorizes employees into Junior, Mid, Senior, and Expert groups
TenureGroup	Groups employees based on years spent in the company
PromotionRate	Measures employee progression based on years at the company relative to time since last promotion
HighIncome	Indicates whether an employee earns above the company median salary

Employee Risk Profile

The employees most likely to leave typically share several characteristics:

- Work overtime
- Frequently travel
- Earn relatively lower salaries
- Are in the early stages of their careers
- Have lower job satisfaction
- Are single

Conversely, employees with longer tenure, higher salaries, no overtime, and greater job satisfaction demonstrate significantly higher retention.

Business Recommendations

1. Reduce Excessive Overtime

Monitor overtime continuously and introduce flexible scheduling, workload balancing, or compensatory leave to reduce burnout.

2. Strengthen Early Employee Retention

Implement structured onboarding programs together with regular follow-up meetings during employees' first years.

3. Review Compensation Strategy

Benchmark salaries against market standards, particularly for junior employees, to ensure competitive compensation.

4. Optimize Business Travel

Rotate travel responsibilities whenever possible and monitor employees who travel frequently to reduce fatigue.

5. Improve Employee Experience

Increase investment in employee engagement initiatives, career development, recognition programs, and professional training.

Future Work

The cleaned dataset is now suitable for predictive machine learning.

Potential models include:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

A predictive attrition model can help HR proactively identify employees at high risk of resignation and intervene before turnover occurs.